INCH-POUND

MIL-P-70543B(AR)
w/AMENDMENT 4
2 February 2004

SUPERSEDING
MIL-P-70543B(AR)
w/AMENDMENT 3
3 December 2001

MILITARY SPECIFICATION

PROJECTILE, 81MM, ILLUMINATING, M853A1
PROJECTILE, 81MM, INFRARED ILLUMINATING, XM816
METAL PARTS

MIL-P-70543B (AR) was inactivated after 3 January 1997 for new design.

This specification is approved for use by the U.S. Army Armament, Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, examinations and tests for metal parts and sub-assemblies for the projectile, 81mm, Illuminating, M853A1 and Infrared, Illuminating, XM816.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN:AMSRD-AAR-AIC-S, Picatinny Arsenal, NJ 07806-5000. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A FSC 1315

DISTRTBUTTON STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

STANDARDS

* FEDERAL

TT-C-490 - Cleaning Methods for Ferrous Surfaces and Pretreatment for Organic Coating

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1169 -Packaging, Packing, and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9381093 - Projectile,81MM: Illuminating, M853A1 Loading Assembly
12953388 - Projectile, 81MM: Infrared Illuminating, XM8I6 Loading Assembly

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

* ASTM B117 Standard Test Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B 46.1 - Surface Texture

(Application for copies should be addressed to: American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings, specifications, and this specification.

3.2 Components and assemblies. The components and assemblies shall comply with the applicable requirements specified on Drawing (dwg.) 9381093 and 12953388 and associated drawings and with all requirements specified in applicable specifications, standards, and this specification.

3.3 First article, When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

3.4 Wood insert. The moisture content of the wood insert shall not exceed six percent for use on the XM816 Infrared Illuminating Cartridge.

3.5 Workmanship. Threads shall be full and undamaged for the entire minimum length or depth. Threads and other parts shall be free of burrs or foreign matter which may interfere with further assembly or proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated with residue. Wood insert for the XM816 Canister loading shall be free of cracks, holes, cavities or other foreign matter.

* 3.6 Salt spray (fog) test. The components and assemblies shall withstand the salt spray requirement cited on drawing 9349905 and 9327836.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as required per Table I.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see TABLE I) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

TABLE I First article inspection
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Projectile, 81MM: Illuminating M853A1 Assembly and Proj, 81MM: Infrared Illum XM816 Assembly	SHEET 1 OF 2		DRAWING NUMBER See below
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY NA INSPECTION METHOD REFERENCE
	Cage, Bearing (Dwg. 9381087) Examination for defects	15	3.2	4.4.2.1
	Canister, Illuminant (Dwg. 9381076) Examination for defects	15	3.2	4.4.2.2
	Plate, Base (Dwg. 9381086) Examination for defects	15	3.2	4.4.2.3
	Pin (Dwg. 9381088) Examination for defects	15	3.2	4.4.2.4
	Sleeve, Split (Dwg. 9369959) Examination for defects	15	3.2	4.4.2.7
	Plate, Dorque (Dwg. 9369956) Examination for defects	15	3.2	4.4.2.8
	Plate, Support (Dwg. 9381085) Examination for defects	15	3.2	4.4.2.10
	Tube, Body (Dwg. 9381090) Examination for defects	15	3.2	4.4.2.11
	Adapter (Dwg. 9381091) Examination for defects	15	3.2	4.4.2.12
NOTES:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

TABLE I First article inspection
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Projectile, 81MM: Illuminating M853A1 Assembly and Proj, 81MM: Infrared Illum XM816 Assembly	SHEET 2 OF 2		DRAWING NUMBER See below
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY NA INSPECTION METHOD REFERENCE
	Body Tube Assembly (Dwg. 9349905) Examination for defects	15	3.2	4.4.2.13
	Cone, Tail (Dwg. 9327836) Examination for defects	15	3.2	4.4.2.14
	Parachute, Flat Circular (Dwg. 9381631) Examination for defects	15	3.2	4.4.2.15
	Parachute Folding and Packing Assembly (Dwg. 9381075) Examination for defects	15	3.2	4.4.2.16
	Bag, Deployment (Dwg. 9381589) Examination for defects	15	3.2	4.4.2.17
	Swivel Assembly (Dwg. 9381083) Examination for defects	15	3.2	4.4.2.18
	Ignition Disc Assembly (Dwg. 9381078) Examination for defects	15	3.2	4.4.2.19
	Canister Illuminant Assembly (Dwg. 9381077) Examination for defects	15	3.2	4.4.2.20
	Wire, Suspension (Dwg. 9381073) Examination for defects	15	3.2	4.4.2.21
*	Salt Spray (Fog) and Adhesion Test Examination for defects	5	3.6	4.5.6
NOTES:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classifications of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE II. Attributes sampling inspection

Lot Size	Inspection Levels					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, variables sampling or other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.4). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.1	TITLE: Cage, Bearing	SHEET 1 OF 1		DRAWING NUMBER 9381087
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381083 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness of plate	Level III	3.2	Gage
102	Diameter of cage	Level III	3.2	Gage
103	Diameter of well	Level III	3.2	Gage
104	Diameter of thru-hole	Level III	3.2	Gage
105	Depth of well	Level III	3.5	Gage
Minor				
201	Radius, incorrect	Level V	3.2	Gage
202	Surface finish improper	Level V	3.2	Visual/4.5.4
203	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

PARAGRAPH 4.4.2.2	TITLE: Canister, Illuminant	SHEET 1 OF 1		DRAWING NUMBER 9381076
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381077 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Wall thickness, min.	Level III	3.2	Gage
102	Diameter of thru-hole,max.	Level III	3.2	Gage
103	Inside diameter of canister, min.	Level III	3.2	Gage
104	Length of canister	Level III	3.2	Gage
105	Hardness	Level III	3.5	4.5.1
Minor				
201	Protective finish missing or Inadequate	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.3	TITLE: Plate, Base	SHEET 1 OF 1		DRAWING NUMBER 9381086
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381083 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter of plate	Level III	3.2	Gage
102	Thickness of plate	Level III	3.2	Gage
Minor				
201	Surface finish improper	Level V	3.2	Visual/4.5.4
202	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

PARAGRAPH 4.4.2.4	TITLE: Pin	SHEET 1 OF 1		DRAWING NUMBER 9381088
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381083 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter above base	Level III	3.2	Gage
102	Height of base	Level III	3.2	Gage
Minor				
201	Outside diameter of base of pin	Level V	3.2	Gage
202	Location of hole on center pin	Level V	3.2	Gage
203	Distance from top to center of hole, min.	Level V	3.2	Gage
204	Diameter of thru-hole, min.	Level V	3.2	Gage
205	Overall height	Level V	3.2	Gage
206	Surface finish improper	Level V	3.2	Visual/4.5.4
207	Inside radius, max.	Level V	3.2	Gage
208	Evidence of poor workmanship	Level V	3.5	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.5	TITLE: Ring, Filler	SHEET 1 OF 1		DRAWING NUMBER 9381080
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381078 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter	Level III	3.2	Gage
102	Diameter of hole	Level III	3.2	Gage
103	Thickness	Level III	3.2	Gage
Minor				
201	Hole missing	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

PARAGRAPH 4.4.2.6	TITLE: Spacer, Felt	SHEET 1 OF 1		DRAWING NUMBER 9381074
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381093/12953388 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Hole missing	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.7	TITLE: Sleeve, Split	SHEET 1 OF 1		DRAWING NUMBER 9369959
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381093/12953388 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Length of sleeve	Level III	3.2	Gage
102	Thickness of sleeve, max.	Level III	3.2	Gage
Minor				
201	Outside radius of sleeve	Level V	3.2	Gage
202	Arc length	Level V	3.2	Gage
203	Perpendicularity of ends	Level V	3.2	Gage
204	Surface finish improper	Level V	3.2	Visual/4.5.4
205	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

PARAGRAPH 4.4.2.8	TITLE: Plate, Droque	SHEET 1 OF 1		DRAWING NUMBER 9369956
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381093/12953388 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter of plate, max.	Level III	3.2	Gage
102	Thickness of plate, min.	Level III	3.2	Gage
Minor				
201	Diameter of thru-hole, min.	Level V	3.2	Gage
202	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.9	TITLE: Cup, Parachute	SHEET 1 OF 1		DRAWING NUMBER 9369955
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381093/12953388 INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Inside radius of cutout	Level V	3.2	Gage
202	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

PARAGRAPH 4.4.2.10	TITLE: Plate, Support	SHEET 1 OF 1		DRAWING NUMBER 9381085
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381077 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter, max.	Level III	3.2	Gage
102	Inside diameter, min.	Level III	3.2	Gage
103	Thickness	Level III	3.2	Gage
104	Flatness	Level III	3.2	Gage
105	Hardness	Level III	3.2	Gage/4.5.2
Minor				
201	Surface finish	Level V	3.2	Visual/4.5.4
202	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.11	TITLE: Tube Body	SHEET 1 OF 1		DRAWING NUMBER 9381090
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9349905 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material damaged or defective	Level III	3.1	Visual
102	Forward outside diameter	Level III	3.2	Gage
103	Rear outside diameter	Level III	3.2	Gage
104	Rear inside diameter	Level III	3.2	Gage
105	Forward inside diameter	Level III	3.2	Gage
106	Central inside diameter	Level III	3.2	Gage
107	Depth of tail cone well	Level III	3.2	Gage
108	Depth of fuze adapter well	Level III	3.2	Gage
109	Distance from front face to start of rear O.D., max.	Level III	3.2	Gage
110	Overall length	Level III	3.2	Gage
111	Runout of rear inside diameter with front face	Level III	3.2	Gage
112	Runout of rear inside diameter to rear face	Level III	3.2	Gage
113	Runout of forward inside diameter with rear inside diameter and rear face	Level III	3.2	Gage
114	Runout of forward outside diameter with rear inside diameter and rear face	Level III	3.2	Gage
115	Runout of rear inside diameter with rear outside diameter	Level III	3.2	Gage
Minor				
201	Surface finish improper	Level V	3.2	Visual/4.5.4
202	Angle rear of forward bourrelet	Level V	3.2	Gage
203	Incorrect chamfer at front face	Level V	3.2	Gage
204	Presence of radius at rear end	Level V	3.2	Visual
205	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.12	TITLE: Adapter	SHEET 1 OF 1		DRAWING NUMBER 9381091
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9349905 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Forward outside diameter	Level III	3.2	Gage
102	Rear outside diameter	Level III	3.2	Gage
103	Runout of front diameter with rear diameter and front shoulder	Level III	3.2	Gage
104	Perpendicularity of shoulder to rear outside diameter	Level III	3.2	Gage
105	Parallelism of front face with Shoulder	Level III	3.2	Gage
106	Pitch diameter of threads	Level III	3.2	Gage
107	Minor diameter of threads, max.	Level III	3.2	Gage
108	Distance from front face to Shoulder	Level III	3.2	Gage
109	Overall length	Level III	3.2	Gage
Minor				
201	Diameter at front of forward Chamfer	Level V	3.2	Gage
202	Diameter at rear of rear chamfer	Level V	3.2	Gage
203	Radius of chamfer missing or Incorrect	Level V	3.2	Visual
204	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.13	TITLE: Body Tube Assembly	SHEET 1 OF 1		DRAWING NUMBER 9349905
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381093/12953388 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Sealant missing or inadequate on mating surfaces	Level III	3.2	Visual
102	Diameter of retaining pin holes	Level III	3.2	Gage
103	Depth of retaining pin holes	Level III	3.2	Gage
104	Retaining pin(s) missing	Level III	3.2	Visual
105	Retaining pen(s) not flush or below	Level III	3.2	Visual/Tactile
106	Length over maximum	Level III	3.2	Gage
107	Excessive gap between body tube and adapter	Level III	3.2	Gage
108	Runout of thread pitch diameter at projected length to rear inside diameter and rear and front faces	Level III	3.2	Gage
109	Diameter over max, after paint	Level III	3.2	Gage
110	Protective finish present on rear inside diameter (I.D.) or adapter threads	Level III	3.2	Visual
111	Corrosion preventative compound missing or not full on threads	Level III	3.2	Visual
* 112	Failure of salt spray (fog) and adhesion test	See 4.5.6	3.6	4.5.6
Minor				
201	Marking incorrect, missing or illegible	Level V	3.2	Visual
202	Protective finish missing or not full	Level V	3.2	Visual
203	Distance of retaining pin holes to face of adapter	Level V	3.2	Gage
204	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.14	TITLE: Cone, Tail	SHEET 1 OF 2		DRAWING NUMBER 9327836
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381093/12953388 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Runout of rear thread at projected distance	Level III	3.2	Gage
102	Perpendicularity of rear face with pitch diameter (PD) of rear thread	Level III	3.2	Gage
103	Runout at rear of taper	Level III	3.2	Gage
104	Runout at front of taper	Level III	3.2	Gage
105	Runout of obturator groove	Level III	3.2	Gage
106	Runout of bourrelet	Level III	3.2	Gage
107	Runout of forward inside diameter	Level III	3.2	Gage
108	Wall thickness	Level III	3.2	Gage
109	Obturator groove diameter	Level III	3.2	Gage
110	Bourrelet diameter after paint, max.	Level III	3.2	Gage
111	Forward ID	Level III	3.2	Gage
112	Pitch diameter of rear thread	Level III	3.2	Gage
113	Minor diameter of rear thread, max.	Level III	3.2	Gage
114	Distance from front face to shoulder	Level III	3.2	Gage
115	Forward OD	Level III	3.2	Gage
116	Overall length	Level III	3.2	Gage
117	Distance from front face to bourrelet	Level III	3.2	Gage
118	Contour of obturator groove (See Note 1)	Level III	3.2	Gage
119	Runout of forward shoulder	Level III	3.2	Gage
* 120	Failure of salt spray (fog) and adhesion test	See 4.5.6	3.6	4.5.6
Minor				
201	Distance from front fact to forward taper diameter	Level V	3.2	Gage
202	Distance from front face to rear taper diameter	Level V	3.2	Gage
203	Protective finish improper	Level V	3.2	Visual
204	Radius incorrect or blends discontinuous	Level V	3.2	Visual
205	Chamfer incorrect	Level V	3.2	Gage
206	Surface finish improper	Level V	3.2	Visual/4.5.4
207	Distance from rear face to base of cavity, min.	Level V	3.2	Gage
208	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.15	TITLE: Parachute, Flat Circular	SHEET 1 OF 1		DRAWING NUMBER 9381631
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381075 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Hole or tear in canopy greater than 1/8 inches	Level III	3.2	Gage
102	One or more shroud line(s) missing	Level III	3.2	Visual
103	Stitching missing or broken	Level III	3.2	Visual
104	Loop missing or improperly taped	Level III	3.2	Visual
105	Free length of vent lines (2 per chute)	Level III	3.2	Gage
106	Missing reinforcing tape	Level III	3.2	Visual
107	More than one patch	Level III	3.2	Visual
108	The patched hole or tear exceeds 1/2 inch max.	Level III	3.2	Gage
109	Location bartack	Level III	3.2	Gage
110	Tape to hold five sets of shroud lines together missing	Level III	3.2	Visual
111	Length of stitched shroud lines together to form a loop	Level III	3.2	Gage
112	Parachute OD	Level III 1/	3.2	Gage
Minor				
201	Length of shroud line at attachment hem, min.	Level V	3.2	Gage
202	Length of reinforcing tape at shroud line attachment hem, min.	Level V	3.2	Gage
203	Length of shroud lines from canopy hem to confluence point	Level V	3.2	Gage
204	Evidence of poor workmanship	Level V	3.5	Visual

NOTES: 1/ OD measurement should be taken using the braided nylon flat tape that is parallel or perpendicular to the straight of the fabric with 2 lbs nominal tension.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.16	TITLE: Parachute Folding and Packing Assembly	SHEET 1 OF 1		DRAWING NUMBER 9381075
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381093/12953388 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Parachute bag stitching missing or broken	Level III	3.2	Visual
102	Parachute bag pull cord missing or improperly attached	Level III	3.2	Visual
103	Improper accordion fold	Level III	3.2	Visual
104	Shroud lines twisted or tangled or 1/2 of shroud lines not coiled under the last fold	Level III	3.2	Visual
105	Length of shroud lines above the parachute bag incorrect	Level III	3.2	Gage
Minor				
201	String missing or improperly tied	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual

PARAGRAPH 4.4.2.17	TITLE: Bag, Deployment	SHEET 1 OF 1		DRAWING NUMBER 9381589
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381075 INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Inside diameter of bag, min.	Level V	3.2	Gage
202	Inside depth of bag, min.	Level V	3.2	Gage
203	Stitching incorrect or broken	Level V	3.2	Visual
204	Evidence of poor workmanship	Level V	3.5	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.18	TITLE: Swivel Assembly	SHEET 1 OF 1		DRAWING NUMBER 9381083
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381084 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Swivel pin fails to rotate freely	Level III	3.2	Visual/Manual
102	Swivel assembly fails to pass through a 2.633 diameter ring gage	Level III	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

PARAGRAPH 4.4.2.19	TITLE: Ignition Disc Assembly	SHEET 1 OF 1		DRAWING NUMBER 9381078
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381079/12953387 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Missing gauze	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual
NOTES:				

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.20	TITLE: Canister Illuminant Assembly	SHEET 1 OF 1		DRAWING NUMBER 9381077
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381094 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Canister illuminating assembly fails to pass through the ring gage	Level III	3.2	Gage
102	Spot weld improper or missing	Level III 1/	3.2	Visual/Manual, 4.5.3
Minor				
201	Length	Level V	3.2	Gage
202	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:1/ In addition to the visual inspection, five (5) parts from each lot shall be tested for adequate weld strength per paragraph 4.5.3.

PARAGRAPH 4.4.2.21	TITLE: Suspension Wire	SHEET 1 OF 1		DRAWING NUMBER 9381073
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 9381075 / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Fails to withstand the minimum breaking strength	1/	3.2	4.5.5
102	Length of wire	Level III	3.2	Gage
Minor				
201	Wire ends not fused	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:1/ Ten (10) parts from each lot shall be tested per 4.5.5. If one or more samples fail to meet the requirement, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

4.5 Testing.

4.5.1 Hardness test of illuminant canister. Hardness reading shall be taken on the outside surface, approximately one inch from the open end.

4.5.2 Hardness test of support plate. Hardness reading shall be taken on the outside surface between the spot welds.

4.5.3 Test for adequate weld strength. Test for adequate weld strength shall be conducted by pulling the plate, support from the canister. No more than one undersize weld button is permitted on each sample tested. More than one weld button which measures less than 0.125 inches diameter shall be cause for rejection of the lot.

4.5.4 Surface roughness. The roughness comparison specimens prescribed by ANSI -B46.1 shall be used as a basis for surface roughness determination. Four (4) power magnification may be used in performing visual examination.

4.5.5 Pull test of suspension wire. The test sample shall be assembled on a universal testing machine and the load applied in an axial direction until the, required pull force is exerted. The rate of travel of the machine shall be 2 inches per minute maximum.

* 4.5.6 Salt spray and adhesion test.

4.5.6.1 Salt spray test sample selection. In performing the following test, TT-C-490 shall apply unless specified otherwise herein. The test shall perform for the First Article inspection, each day of painting during production, and whenever a change is made to the paint process, until all samples from five (5) consecutive days of painting have passed the test. Thereafter, the sample frequency may be reduced to twice per week during continuous days of painting. Five (5) body tube assemblies and five (5) tail cones shall be selected, consisting of randomly chosen body tube assemblies and tail cones which have been painted at the same time using the same equipment as other body tubes and tail cones. The samples shall be taken from each continuous final paint operation up to 24 hours long. One (1) body tube assembly and one (1) tail cone shall be taken during the first thirty (30) minutes of painting, one (1) body tube assembly and one (1) tail cone during the last thirty (30) minutes and the others at any other time.

Prior to salt spray testing, two (2) body tube assemblies and two (2) tail cones must only pass an adhesion test using the procedures and pass-fail criteria contained in 4.2.8 and 3.5.8 respectively, of TT-C-490. One (1) of the adhesion tests of the body tube assembly shall be performed along the circumference of the fuze adapter and body tube interface. During this adhesion test, the edge of the tape shall be laid adjacent to the front face such that the side of the adapter is fully covered. The second adhesion test shall be performed in the usual fashion along the length of the body tube

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

assembly. If both body tube assemblies and tail cones pass the adhesion test, the three (3) body tubes and three (3) tail cones remaining shall be scored lengthwise using a precision scriber tool/insert or equivalent in preparation for the salt spray test.

4.5.6.2 Salt spray test. The salt solution, test equipment and drying procedures shall be in accordance with ASTM B117, Salt Spray (Fog) Testing, unless specified otherwise herein. Specimens shall be positioned between 15 to 30 degrees from vertical during the test. No coating of any kind shall be permitted over the paint on stamp marks of the specimens prior to testing, however, areas in contact with racks or supports, unpainted surfaces and areas within 1/8 inch of unpainted surfaces may be protected.

When examining the specimens for signs of paint failure, the significant surface shall be 60 degrees on either side of a scribe line drawn down the length of the specimens, except in the obturator groove and within 1/8 inch from any edges.

As used herein, the term “blister” refers to an unbroken area of paint with a bubble-like appearance whereas “pits” or “spots” are exposed chromate phosphate pretreatment or bare metal visible through holes in the paint. Blisters which are present when the test is over are to be counted within five minutes of removal from the test chamber.

After testing, any body tube assembly or tail cone that fails for any of the following reasons shall be cause for rejection of all applicable parts painted since the last successful test, and a return to the original sampling plan if the failure occurred during reduced sampling. The failure criteria in a thru i applies to the significant surface only.

a. The paint shows more than 1/8 inch creepage, blistering, or loss of adhesion of paint from the scribe mark, stamp mark, or edge.
b. There are more than five (5) blisters total.
c. Any blister exceeding 1/16 inch average diameter.
d. Any pit exceeding 1/8 inch average diameter.
e. There are four (4) or more scattered pits exceeding 1/16 inch average diameter.
f. There are three (3) or more spots within one (1) inch.
g. There are two (2) spots within one (1) square inch in three (3) or more places.
h. There are more than ten (10) scattered spots total.
i. There is evidence of corrosion under the paint (a discoloration showing through the paint) in excess of ¼ inch in diameter total area.
j. There is some variation of the above that the Government representative deems excessive, or an unlisted defect such as inadequate paint coverage anywhere on the body, including the obturating band groove.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

4.6 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the “Inspection Method Reference” column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

5. PACKAGING

5.1 Packaging , packing, and marking. Packaging, packing, and marking shall be in accordance with the requirements of level C of MIL-STD-1169.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for assembly to M853A1 Illuminating and XM816 Infrared, illuminating cartridges for use on the 81mm mortar.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-C, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Subject term (key word) listing.

Canister
Cartridge
Illuminant
Strength, weld

6.7 Amendment notations. The margins of this amendment are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 13l5-0276)

Note: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at www.dodssp.daps.mil

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:39Z
Check the source to verify that this is the current version before use.